

Lecture 7 – Virtualization (25 MCQs)

1. Virtualization is important mainly because:

- A) Software changes faster than hardware
- B) Hardware changes faster than software
- C) It eliminates the need for operating systems
- D) It removes the need for networking

Answer: B

2. One major benefit of virtualization is that it:

- A) Increases CPU speed
- B) Makes software dependent on hardware
- C) Improves portability and code migration
- D) Eliminates isolation

Answer: C

3. Isolation in virtualization means that:

- A) Virtual machines share memory
- B) Failures in one VM do not affect others
- C) All VMs run the same OS
- D) Applications directly access hardware

Answer: B

4. The core principle behind virtualization is:

- A) Centralized control
- B) Direct hardware access
- C) Mimicking interfaces
- D) Eliminating system calls

Answer: C

5. In interface mimicking, software believes it is interacting with:

- A) A network
- B) A virtual machine monitor
- C) Real hardware or native interfaces
- D) Another application

Answer: C

6. Which interface level defines the machine instructions of a CPU?

- A) API
- B) System calls
- C) Instruction Set Architecture (ISA)
- D) Middleware

Answer: C

7. Privileged instructions are:

- A) Allowed for all applications
- B) Executed only by user programs
- C) Allowed only in kernel mode
- D) Used only in virtualization

Answer: C

8. System calls exist mainly to:

- A) Improve performance
- B) Allow applications to safely request OS services
- C) Replace APIs
- D) Virtualize hardware

Answer: B

9. Library calls (APIs) primarily exist to:

- A) Control hardware
- B) Replace system calls
- C) Simplify programming and improve portability
- D) Manage CPU scheduling

Answer: C

10. A Process Virtual Machine (Process VM):

- A) Runs directly on hardware
- B) Virtualizes an entire operating system
- C) Virtualizes a program execution environment
- D) Requires a hypervisor

Answer: C

11. Which of the following is an example of a Process VM?

- A) VMware ESXi
- B) Hyper-V
- C) Java Virtual Machine (JVM)
- D) Xen

Answer: C

12. Native Virtual Machine Monitors (VMMs) are characterized by:

- A) Running on top of a host OS
- B) Running directly on hardware
- C) Low isolation
- D) High overhead

Answer: B

13. Which virtualization approach provides the best performance?

- A) Process VM
- B) Hosted VMM
- C) Native VMM
- D) Container

Answer: C

14. Hosted VMMs differ from native VMMs because they:

- A) Do not support full OSs
- B) Run as applications on a host OS
- C) Run directly on hardware
- D) Eliminate virtualization overhead

Answer: B

15. Containers differ from virtual machines because containers:

- A) Run a full guest OS
- B) Do not provide isolation
- C) Share the host OS kernel
- D) Cannot limit resources

Answer: C

16. Which kernel feature provides isolated views of system resources in containers?

- A) Hypervisors
- B) Virtual disks
- C) Namespaces
- D) APIs

Answer: C

17. Control groups (cgroups) are used to:

- A) Isolate file systems
- B) Virtualize hardware
- C) Restrict CPU and memory usage
- D) Replace namespaces

Answer: C

18. Union file systems in containers are mainly used to:

- A) Encrypt files
- B) Allow multiple containers to share read-only layers
- C) Improve CPU scheduling
- D) Replace storage devices

Answer: B

19. Compared to virtual machines, containers are:

- A) Heavier and slower
- B) More isolated
- C) Lightweight with faster startup
- D) Dependent on hypervisors

Answer: C

20. A key limitation of containers is that they:

- A) Cannot scale
- B) Cannot isolate processes
- C) Must share the host OS type
- D) Cannot run applications

Answer: C

21. PlanetLab is best described as:

- A) A centralized cloud provider
- B) A peer-to-peer system
- C) A global distributed testbed
- D) A single virtual machine

Answer: C

22. PlanetLab uses virtualization mainly to:

- A) Improve performance
- B) Reduce network latency
- C) Prevent experiments from interfering with each other
- D) Eliminate hardware failures

Answer: C

23. A VServer in PlanetLab is:

- A) A full virtual machine
- B) A lightweight isolated environment
- C) A physical server
- D) A kernel module only

Answer: B

24. Infrastructure-as-a-Service (IaaS) provides users with:

- A) Only applications
- B) Runtime environments only
- C) Virtual machines and basic infrastructure
- D) Fully managed software

Answer: C

25. In Software-as-a-Service (SaaS), the user:

- A) Manages the OS and runtime
- B) Manages application code

- C) Manages nothing
- D) Manages virtual machines

Answer: C